How can we compute this likelihood? GPyTorch offers a convenient class `gpytorch.mlls.ExactMarginalLogLikelihood` that takes in a GP model and computes the marginal log likelihood of its training data given the hyperparameters of the model.

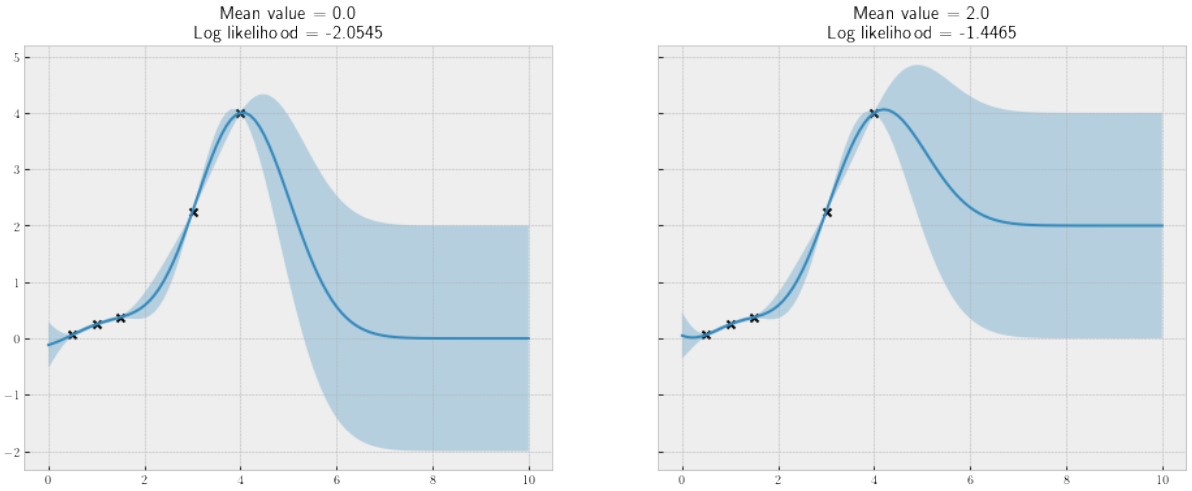


Figure 3.6 GP predictions given two different constant mean functions. The value 2 gives a higher likelihood value than the value 0, indicating that the former mean function is a better fit than the latter mean function.

To see that this likelihood quantity is effective at quantifying data fit, we consider Figure 3.6. This figure visualizes the predictions made by two separate GP models: a zero mean GP that we have seen in the previous subsection on the left, and the GP with a mean function whose value is 2 on the right. Notice how in the second panel, the mean function reverts to 2 instead of 0 on the right side of the plot. Here, the second GP has a higher (log) likelihood than the first, which means the value 2 explains our training data better than the value 0.

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